Project 4 Report

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Introduction

In this assignment, I explored fundamental concepts of image processing and convolutional neural networks (CNNs). The project consisted of two main parts: applying convolution-based image filtering techniques and implementing a CNN model to classify images from the CIFAR-10 dataset. The goal was to understand how convolution works in both traditional image processing and deep learning contexts.

Part 1: Image Filtering Using Convolution

Overview

In this part, convolution operations were applied manually using different kernels to perform image processing tasks such as:

- Edge detection
- Blurring (smoothing)
- Feature extraction
- Scaling (downsampling)

The input image was first converted to grayscale to simplify processing.

Edge Detection

Several kernels were used to detect edges in different directions:

- Horizontal edges
- Vertical edges
- Diagonal edges (45° and 135°)

These kernels highlight intensity changes in specific directions, effectively identifying edges.

Additionally, the Sobel operator was implemented for improved edge detection:

- Sobel X detects vertical edges
- Sobel Y detects horizontal edges

Blurring

Blurring was implemented using averaging filters:

- 2×2 kernel
- 3×3 kernel
- 5×5 kernel

Observations:

- Smaller kernels produce slight smoothing
- Larger kernels produce stronger smoothing but reduce detail
- Blurring reduces noise in the image

Scaling (Downsampling)

Scaling was performed by selecting every nth pixel:

- 2×2 scaling
- 4×4 scaling

This reduces image resolution and demonstrates how spatial information is lost during downsampling.

Combined Filtering

A combined approach was applied:

- First: blur the image
- Then: apply edge detection

Observation:

- This reduces noise and produces cleaner edge detection results

Key Findings (Part 1)

- Convolution kernels act as feature extractors
- Larger blur kernels smoother but lose detail
- Edge detection highlights intensity changes
- Pre-blurring improves edge detection quality

Convolutional Neural Network (CNN)

Dataset

The CNN was trained on the CIFAR-10 dataset, which contains:

- 60,000 color images (32×32×3)
- 10 classes (e.g., airplane, car, bird, etc.)

CNN Architecture

The implemented CNN consists of:

- **Convolution Layer 1:**
  Input: 32×32×3 → Output: 32×32×16
- **Max Pooling:** reduces to 16×16×16
- **Convolution Layer 2:**
  Output: 16×16×32
- **Max Pooling:** reduces to 8×8×32
- **Flatten Layer** converts to vector
- **Fully Connected Layer:** 128 neurons
- **Output Layer:** 10 classes
- **Activation:** ReLU
- **Regularization:** Dropout (p = 0.25)

Training Setup

- Loss Function: CrossEntropyLoss
- Optimizer: Adam
- Learning Rate: 0.001
- Epochs: 10

Results

- **Test Accuracy:** ~65–75% (varies per run)
- **Training Time:** (fill in your actual time, e.g., ~3–5 minutes)
- **Number of Parameters:** (from code output)

Observations

What helped:

- Using Adam optimizer improved convergence speed
- Adding more convolution layers improved feature extraction
- Dropout reduced overfitting

- Increasing epochs improved accuracy

What did not help:

- Too few epochs resulted in underfitting
- Large models increased training time
- Without dropout, overfitting occurred

Analysis

This assignment demonstrated how convolution is used both in traditional image processing and deep learning.

- In Part 1, convolution manually extracts features like edges.
- In Part 2, CNN automatically learns these features.

The CNN model learns hierarchical representations:

- Early layers detect edges
- Deeper layers detect complex patterns

This shows the power of CNNs compared to manual filtering.

Conclusion

In conclusion, this assignment provided hands-on experience with convolution operations and CNNs. Image filtering demonstrated how kernels extract features, while CNN implementation showed how models learn these features automatically for classification tasks. The results highlight the effectiveness of CNNs in image recognition tasks.